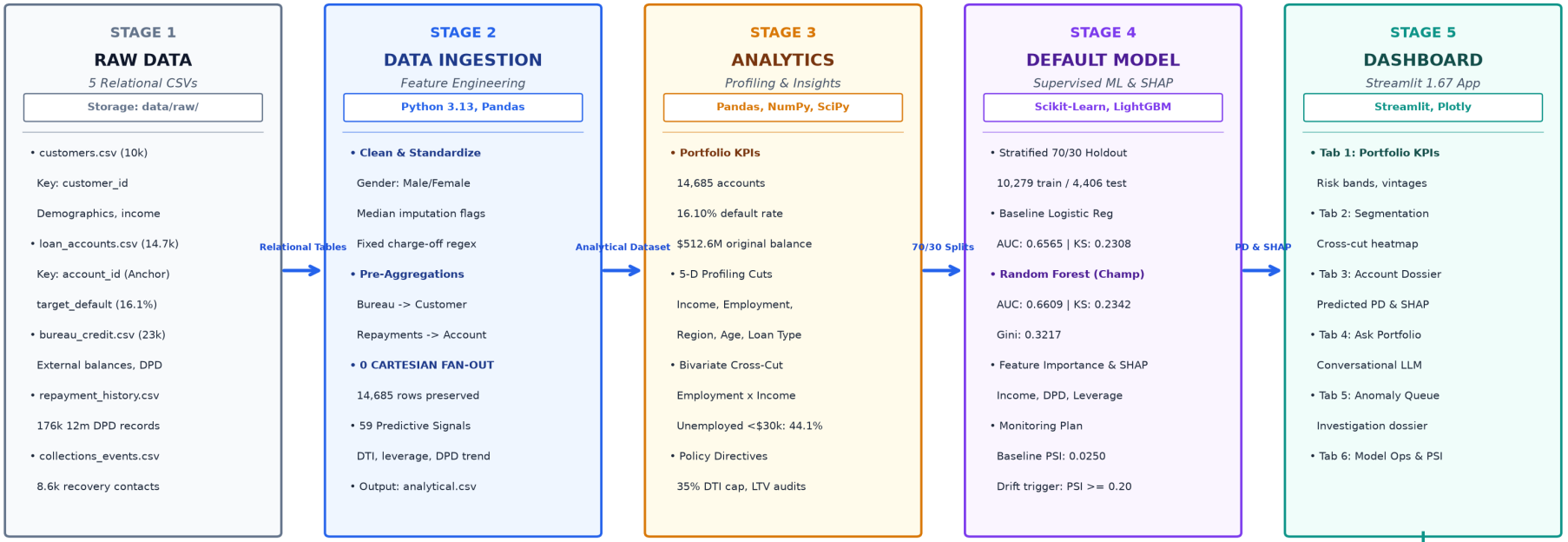
# CREDIT RISK PORTFOLIO MONITORING PLATFORM — SYSTEM ARCHITECTURE

End-to-End Pipeline: Raw Data → Ingestion → Analytics → Default Model → Dashboard → Dual Embedded AI Systems

*Tool/Library per Stage | Exact LLM Calling Endpoint & Integration Point | Strict Zero-PII Aggregate Payload Schema*




**Query + Filtered Cohort Summary JSON** | **Slice Multidimensional Feature Matrix**

## AI FEATURE 1: 'ASK THE PORTFOLIO'

*Conversational LLM Assistant (Streamlit Tab 4)*

**Groq Cloud API | OpenAI SDK**

- **OPERATIONAL TRIGGER: Risk analyst submits natural language prompt in Streamlit Tab 4.**

### WHERE THE LLM IS CALLED (Execution Point & Endpoint)

- Invoking Function: src/ai_features.py :: query_portfolio_llm()
- Protocol: Outbound TLS 1.3 HTTPS POST request
- Remote Endpoint: https://api.groq.com/openai/v1/chat/completions
- Client SDK: openai.OpenAI / groq.Groq Python Client
- Serving Model: openai/gpt-oss-20b (fallback: gpt-4o-mini)
- Offline Fallback: Deterministic local analytical calculator executes if GROQ_API_KEY / OPENAI_API_KEY is unset or network fails.

### WHAT DATA IS SENT TO IT (Strict Aggregate Evidence Payload)

- Transmitted Data: ONLY Filtered Cohort Summary Evidence JSON:
  - Active Filters: loan_type, region, employment, income_band
  - Slice KPIs: total accounts (N), defaults, default_rate %, balance $
  - Sub-segment Breakdown: default rates by loan type & income band
  - Cross-cut Matrix: delinquency across employment x income
- `Payload Sample: {"filters":{"loan_type":"All"}, "kpis":{"accounts":14685, "defaults":2364, "default_rate":"16.10%"}, "by_product":{...}}`

### ZERO-PII COMPLIANCE & HUMAN-IN-THE-LOOP (HITL)

- ZERO Customer PII (No names, SSNs, phone numbers, or emails transmitted).
- ZERO Account IDs or row-level records sent. Mathematical aggregates only.
- Anti-hallucination prompt citations; unanswerable queries explicitly refused.
- Human-in-the-Loop: Analyst verifies figures against KPI cards before memo use.

## AI FEATURE 2: ANOMALY DETECTION

*Unsupervised Review Queue (Streamlit Tab 5)*

**Scikit-Learn | IsolationForest**

- **OPERATIONAL TRIGGER: Automated background execution on filtered cohort in Tab 5.**

### DETECTION ENGINE & MULTIDIMENSIONAL BENCHMARKING

- Unsupervised Engine: Scikit-Learn IsolationForest (contamination=0.03)
- Multidimensional Feature Matrix: annual_income, dti_ratio, total_debt_to_income, repayment_max_dpd, repayment_mean_dpd, balance
- Peer Group Normalization: Compares borrower against specific cohort benchmarks (Loan Type x Income Band peer medians)
- Trajectory Heuristic: Flags rapid DPD surges (late 6m vs early 6m)
- Severity Ranking: Flagged accounts sorted by anomaly distance score.

### TRANSPARENT REASON ATTRIBUTION & INVESTIGATION DOSSIER

- Explainable Cause Attribution: System generates explicit trigger causes:
  - Delinquency Outlier: 'Peak DPD 60d vs peer group median 8d (+52d)'
  - Bureau Distress: 'External bureau write-offs present (Charge-off flag)'
  - Excessive Leverage: 'DTI 8.4x vs peer group median 2.1x'
  - Trajectory Surge: 'Late 6m DPD +45d higher than early 6m DPD'
- Interactive Investigation Dossier: Side-by-side borrower profile vs peer benchmarks across balance, terms, collections, and repayment curves.

### DECISION AUDIT PERSISTENCE & HUMAN-IN-THE-LOOP MANDATE

- Analyst Sign-Off: Selects Confirm / Dismiss / Escalate & enters notes.
- Persistent Storage: data/processed/analyst_review_decisions.json (Audit Log).
- Regulatory Compliance: Prevents automated credit line freezes without review.
- Fair Lending: Provides compliant adverse action disclosure reasons (FCRA/ECOA).